

HoxB-derived *hoxba* and *hoxbb* clusters are essential for the anterior-posterior positioning of zebrafish pectoral fins

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eLife Assessment

This **important** study advances our understanding of vertebrate forelimb development, specifically the contribution of Hox genes to zebrafish pectoral fin formation. The authors have employed a robust and extensive genetic approach to tackle a key and unresolved question. The findings are overall **convincing** and will be of broad interest to developmental and evolutionary biologists.

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Abstract

Vertebrate paired appendages, such as the pectoral fins in fish and the forelimbs in tetrapods, arise at specific regions along the anterior-posterior axis of the body. Hox genes have long been considered prime candidates for determining the anteroposterior positioning of these paired appendages during development. Evidence from various model organisms, including mouse and chick, supports a role for Hox genes in limb positioning. However, despite extensive phenotypic analyses of numerous single and compound Hox knockout mice, clear genetic evidence for substantial defects in limb positioning has been limited, leaving questions unresolved. In a previous study, we generated seven distinct *hox* cluster-deficient mutants in zebrafish. Here, we provide genetic evidence that zebrafish *hoxba;hoxbb* cluster-deleted mutants specifically exhibit a complete lack of pectoral fins, accompanied by the absence of *tbx5a* expression in pectoral fin buds. In these mutants, *tbx5a* expression in the pectoral fin field of the lateral plate mesoderm fails to be induced at an early stage, suggesting a loss of pectoral fin precursor cells. Furthermore, the competence to respond to retinoic acid is lost in *hoxba;hoxbb* cluster mutants, indicating that *tbx5a* expression cannot be induced in the pectoral fin buds. We further identify *hoxb4a*, *hoxb5a*, and *hoxb5b* as pivotal genes underlying this process. Although the frameshift mutations in these *hox* genes do not recapitulate the absence of pectoral fins, we demonstrate that deletion mutants at these genomic loci show the absence of pectoral fins with low penetrance. Our results suggest that, by establishing the expression domains along the anteroposterior axis, *hoxb4a*, *hoxb5a*,

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Introduction

In jawed vertebrates, paired appendages—such as the pectoral and pelvic fins in fish, and their homologous forelimbs and hindlimbs in tetrapods—develop at precise locations along the anterior-posterior axis of each species. These paired appendages arise from progenitor cells located in distinct regions of the lateral plate mesoderm (Murata et al., 2010 [DOI](#); Nishimoto and Logan, 2016 [DOI](#); Shimada et al., 2013 [DOI](#)). The anteroposterior positioning of these paired appendages has long fascinated researchers, yet our understanding of the molecular mechanisms underlying this process remains limited.

In bilaterian animals, Hox genes—encoding evolutionarily conserved homeodomain-containing transcription factors—provide positional information and developmental timing along the anterior-posterior axis (Iimura and Pourquie, 2007 [DOI](#); Izpisua-Belmonte and Duboule, 1992 [DOI](#); Krumlauf, 1994 [DOI](#)). A defining feature of Hox genes is their structural organization into Hox clusters, where multiple Hox genes are arranged in a precise order. Additionally, Hox clusters exhibit a distinctive phenomenon known as Hox collinearity, where the genomic arrangement of Hox genes correlates with specific developmental regions along the body axes (Dolle et al., 1989 [DOI](#); Duboule and Dolle, 1989 [DOI](#); Graham et al., 1989 [DOI](#)). In vertebrates, Hox clusters underwent divergence due to two rounds of whole-genome duplication early in vertebrate evolution (Dehal and Boore, 2005 [DOI](#); Ohno, 1970 [DOI](#)), leading to the establishment of four distinct Hox clusters (HoxA, HoxB, HoxC, and HoxD), each consisting essentially of 1-13 paralogous groups in tetrapods. In contrast, teleost fishes experienced an additional teleost-specific whole-genome duplication, followed by the loss of a hox cluster, resulting in seven hox clusters in zebrafish (Amores et al., 1998 [DOI](#); Woltering and Durston, 2006 [DOI](#)).

Multiple studies in chick and mouse have suggested that the initial anteroposterior position of limbs is regulated by Hox genes (Tanaka, 2013 [DOI](#); Tickle, 2015 [DOI](#)). For example, the anterior boundaries of Hox gene expression domains were shown to align with future limb positions in

chick embryos (Burke et al., 1995 [↗](#); Cohn et al., 1997 [↗](#)). Experimental manipulations in avian embryos, including overexpression or interference of specific Hox genes, led to altered positions of forelimb buds (Moreau et al., 2019 [↗](#)). In mouse, *Hoxb5* knockout mutants exhibit a rostral shift of forelimb buds with incomplete penetrance (Rancourt et al., 1995 [↗](#)), while other subtle shifts in limb positioning have also been reported in Hox mutants (Royle et al., 2021 [↗](#)). Furthermore, loss of *Gdf11* in mice, which alters the expression of posterior Hox9–13 genes, causes posterior displacement of hindlimb buds, whereas ectopic *Gdf11* induces anterior shifts of hindlimb position (Matsubara et al., 2017 [↗](#)). At the molecular level, Hox proteins directly bind to the Tbx5 limb enhancer and regulate its expression, providing a mechanistic link between Hox activity and forelimb initiation (Minguillon et al., 2012 [↗](#); Nishimoto et al., 2014 [↗](#)). Together, these studies provide strong evidence across different model organisms that Hox genes contribute to the regulation of limb positioning. Nevertheless, despite the generation of numerous single and compound Hox knockout mice, no severe defects in the initial positioning of limb buds have been documented, and the precise mechanisms by which Hox genes specify limb position remain incompletely understood. This stands in contrast to the well-established role of Hox genes in limb patterning after limb bud formation, where paralogous group 9–13 genes in the HoxA and HoxD clusters cooperatively control the proximal–distal axis of developing limbs (Boulet and Capecchi, 2004 [↗](#); Davis et al., 1995 [↗](#); Fromental-Ramain et al., 1996a [↗](#); Fromental-Ramain et al., 1996b [↗](#); Kmita et al., 2005 [↗](#)). Consequently, the exact role of Hox genes in defining the initial position of limb formation along the anterior-posterior axis remains unclear.

Using zebrafish, which belong to the vertebrate class Actinopterygii and are distinct from the tetrapods of the class Sarcopterygii, we previously generated mutants lacking each of the seven zebrafish hox clusters using the CRISPR-Cas9 method (Yamada et al., 2021 [↗](#)). In our recent genetic analyses of *hoxaa*, *hoxab*, and *hoxda* clusters—orthologous to the mouse HoxA and HoxD clusters—we demonstrated that, similar to mice, these zebrafish hox clusters cooperatively play an essential role in the formation of the pectoral fins (Ishizaka et al., 2024 [↗](#)), which are homologous to the forelimbs. In this study, we provide the first genetic evidence, to our knowledge, that Hox genes specify the positions of paired appendages in vertebrates. The double-deletion mutants of *hoxba* and *hoxbb* clusters, derived from the ancient HoxB cluster, exhibit a complete absence of pectoral fins due to the failure to express *tbx5a*. Despite incomplete penetrance, we propose a model in which *hoxb4a*, *hoxb5a*, and *hoxb5b* cooperatively provide positional cues along the anterior-posterior axis within the lateral plate mesoderm, thereby specifying the initial positions for fin bud formation through the induction of *tbx5a* in the pectoral fin field.

Results

Absence of pectoral fins in *hoxba*;*hoxbb* cluster-deleted zebrafish

In a previous study, we created seven individual hox cluster-deficient mutants in zebrafish using the CRISPR-Cas9 system (Yamada et al., 2021 [↗](#)). Among these mutants, *hoxba* cluster-deleted embryos exhibited morphological abnormalities in their pectoral fins at 3 dpf (Figure 1A–C [↗](#)). To further investigate the pectoral fin phenotype in *hoxba* cluster mutants, we first analyzed the expression patterns of *Tbx5* orthologs. Zebrafish possess two paralogs, *tbx5a* and *tbx5b* (Boyle-Anderson et al., 2022 [↗](#)). Among them, *tbx5a* plays a predominant role in the initial induction of pectoral fin buds in zebrafish (Ahn et al., 2002 [↗](#); Garrity et al., 2002 [↗](#); Ng et al., 2002 [↗](#)). When comparing with *tbx5a* expression in wild-type embryos, we found that the *tbx5a* signal was reduced in the pectoral fin buds of *hoxba* cluster mutants (Figure 1G, H [↗](#)). Given that *hoxba* and *hoxbb* clusters originated from the ancestral HoxB cluster through teleost-specific whole-genome duplication (Amores et al., 1998 [↗](#)), there may be functional redundancy between them. Surprisingly, we discovered that the simultaneous deletion of both *hoxba* and *hoxbb* clusters resulted in the complete absence of pectoral fins (Figure 1A, F [↗](#)). In contrast, pectoral fins were present in *hoxba*^{−/−};*hoxbb*^{+/−} and *hoxba*^{+/−};*hoxbb*^{−/−} mutants (Figure 1D, E [↗](#)), indicating that an allele

from either *hoxba* or *hoxbb* cluster is sufficient for the pectoral fin formation. Moreover, all embryos lacking pectoral fins were identified as *hoxba;hoxbb* double homozygous mutants, with the expected penetrance ($n=15/252$; 5.9%) being consistent with predictions based on Mendelian genetics ($1/16=6.3\%$). Furthermore, while *hoxba;hoxbb* cluster double homozygous mutants are embryonic lethal around 5 dpf, we could not detect any trace of pectoral fin development. Alongside this phenotype, the expression of *tbx5a* was significantly reduced to nearly undetectable levels in *hoxba;hoxbb* cluster mutants at 30 hpf (**Figure 1G-L**). The absence of pectoral fins in zebrafish *hoxba;hoxbb* cluster mutants sharply contrasts with the results of a previous study, which showed that mice lacking all HoxB genes except for *Hoxb13* of the HoxB cluster did not exhibit apparent abnormalities in their forelimbs (Medina-Martinez et al., 2000). Our genetic results suggest that the zebrafish *hoxba* and *hoxbb* clusters cooperatively play a crucial role in pectoral fin formation.

Pectoral fin loss specific to *hoxba;hoxbb* cluster-deleted zebrafish

Multiple genetic studies in mice have demonstrated the functional redundancy of Hox genes across the four Hox clusters (Horan et al., 1995; McIntyre et al., 2007; van den Akker et al., 2001; Wellik and Capecchi, 2003; Wellik et al., 2002). To investigate whether the absence of pectoral fins is specific to *hoxba;hoxbb* cluster mutants, we created double-hox cluster mutants by crossing *hoxba* cluster mutants with six other hox cluster mutants in zebrafish. The combinatorial mutations of *hoxba* cluster with five other hox clusters, excluding *hoxbb* cluster, did not enhance the reduced expression of *tbx5a* (**Figure 2A-H**). Furthermore, we examined combined mutants of hox clusters, excluding *hoxba* and *hoxbb*. In *hoxca^{-/-};hoxcb^{-/-}* cluster mutants, where the functions of HoxC genes are completely absent, the expression patterns of *tbx5a* in fin buds were indistinguishable from those of wild-type embryos (**Figure 2I**). Additionally, our previous study showed that *tbx5a* expression is unaffected in *hoxaa^{-/-};hoxab^{-/-};hoxda^{-/-}* cluster mutants (Ishizaka et al., 2024), which lack all HoxA and HoxD-related genes. The results of our deletion mutations of zebrafish hox clusters in various combinations emphasize that the diminished expression of *tbx5a* in the pectoral fin buds is evident only in *hoxba;hoxbb* cluster mutants. Taken together, in contrast to the functional redundancy of the four Hox clusters in mice, our genetic analysis indicates that the Hox clusters responsible for the specification of pectoral fin buds are restricted to the HoxB-derived *hoxba* and *hoxbb* clusters among the seven hox clusters in zebrafish.

Loss of fin progenitors with posterior expansion of cardiac marker expression

To understand the absence of pectoral fin formation in *hoxba;hoxbb* cluster mutants, we examined the expression patterns of *tbx5a* during embryogenesis (**Figure 3A-F**). In wild-type embryos at the 10-somite stage, *tbx5a* expression appeared as bilateral symmetric stripes in the anterior lateral mesoderm, extending from the posterior midbrain to the second somite (**Figure 3A**) (Ahn et al., 2002). These *tbx5a*-positive cells subsequently divided into two groups: the anteriorly migrating cells form the heart primordia, while the posteriorly migrating cells give rise to the future pectoral fin buds (**Figure 3C, E**). In contrast, abnormal expression patterns of *tbx5a* were evident at early stages in *hoxba;hoxbb* cluster mutants. Compared to wild-type embryos, the bilateral stripes of *tbx5a* expression were shortened along the anterior-posterior axis, and the *tbx5a*-positive signal was absent from the posterior region where pectoral fin progenitor cells typically emerge (**Figure 3B**). Although anterior migration toward the presumptive heart occurred, posterior migration toward the fin buds was undetectable, consistent with the loss of pectoral fin progenitor cells in *hoxba;hoxbb* cluster mutants (**Figure 3D, F**). These results underscore the early role of the *hoxba* and *hoxbb* clusters in the formation of pectoral fin buds through the induction of *tbx5a* expression. In zebrafish, studies have shown a reciprocal relationship between pectoral fin and heart progenitor populations in the anterior lateral mesoderm, where retinoic acid signaling promotes pectoral fin formation while restricting cardiac

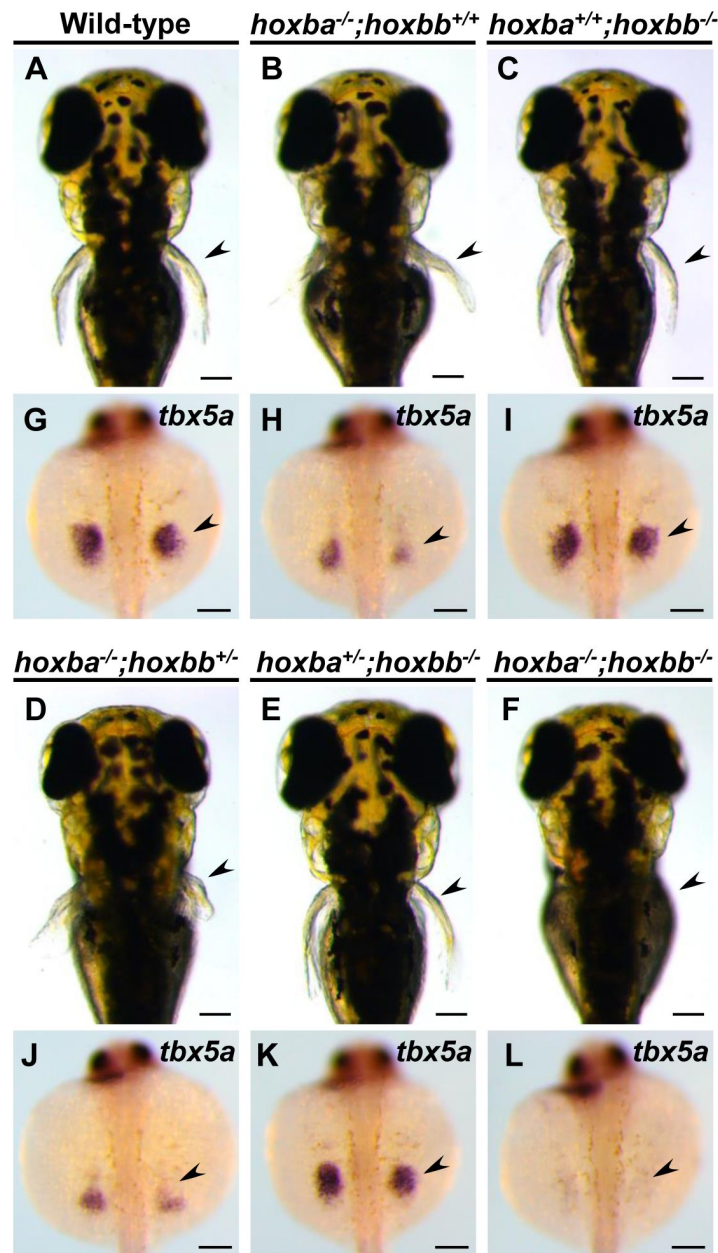


Figure 1.

Lack of pectoral fins in *hoxba*;*hoxbb* cluster-deleted mutants.

(A-F) Dorsal views of live zebrafish larvae at 3 dpf, obtained from intercrosses between *hoxba*;*hoxbb* cluster-deficient hemizygous fish. Arrowheads indicate the positions of the pectoral fins. (G-L) Expression patterns of *tbx5a* in the pectoral fin bud (arrowhead) at 30 hpf. Dorsal views are shown, and the genotype of each specimen was determined. For each genotype, reproducibility was confirmed with at least three different specimens. Genotyping revealed that all the embryos lacking pectoral fins ($n=15$) were *hoxba*;*hoxbb* double homozygotes. Scale bars are 100 μm.

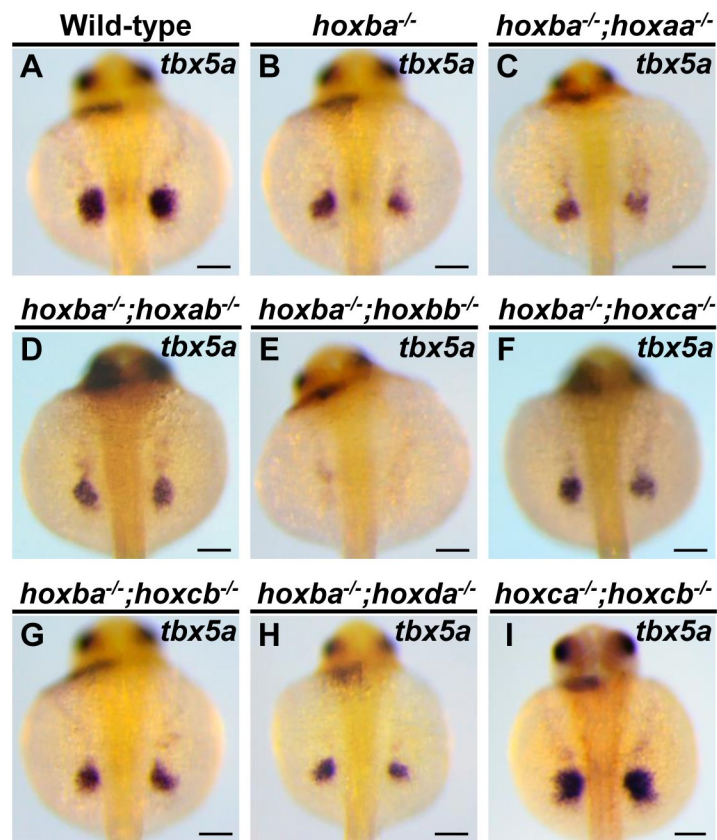


Figure 2.

Significantly decreased expression of *tbx5a* in the pectoral fin buds is specific to *hoxba*/*hoxbb* cluster-deleted mutants.

(A-I) Expression patterns of *tbx5a* in the pectoral fin buds of combinatorial deletion mutants of zebrafish *hox* clusters. Dorsal views of embryos at 30 hpf are displayed. After capturing images, the genotype of each specimen was determined. For each genotype, reproducibility was confirmed with at least three different specimens. Scale bars are 100 μ m.

progenitors (Waxman et al., 2008 [↗](#)). Therefore, we also analyzed cardiac development (**Figure 3G–J** [↗](#)). By examining the expression patterns of *nkx2.5*, an early cardiac progenitor marker (Chen and Fishman, 1996 [↗](#); Serbedzija et al., 1998 [↗](#)), we found that the expression domain of *nkx2.5* in *hoxba;hoxbb* cluster mutants extended more posteriorly compared to wild-type embryos (**Figure 3G, H** [↗](#)). Furthermore, the region positive for the differentiated cardiac cell-specific myosin light chain (*cmlc2*) showed abnormal morphology relative to wild-type embryos (**Figure 3I, J** [↗](#)) (Yelon et al., 1999 [↗](#)). Taken together, these results indicate that *hoxba* and *hoxbb* clusters are essential for establishing the pectoral fin field in zebrafish. In their absence, pectoral fin progenitor cells fail to emerge, accompanied by a posterior expansion of cardiac marker expression domain. Our results highlight a shift in the balance between fin and cardiac progenitor populations within the anterior lateral plate mesoderm.

***hoxba/bb* clusters required for RA-mediated *tbx5a* induction in the fin buds**

Retinoic acid (RA) is a well-known upstream regulator of Hox genes in vertebrates (Boncinelli et al., 1991 [↗](#); Langston and Gudas, 1994 [↗](#); Marshall et al., 1996 [↗](#)). Loss of function of *retinaldehyde dehydrogenase 2* (*raldh2*), an essential regulator in RA synthesis, leads to the absence of pectoral fins in zebrafish (Begemann et al., 2001 [↗](#); Grandel et al., 2002 [↗](#)), a phenotype similar to that observed in *hoxba;hoxbb* cluster mutants. Previous studies have shown that exogenous RA treatments can rescue pectoral fin formation in *raldh2*^{−/−} embryos (Gibert et al., 2006 [↗](#); Grandel et al., 2002 [↗](#)). It has been suggested that RA directly or indirectly induces *tbx5a* expression in zebrafish fin buds (Gibert et al., 2006 [↗](#); Grandel and Brand, 2011 [↗](#); Neto et al., 2012 [↗](#)). The phenotypic similarities between *hoxba;hoxbb* cluster mutants and *raldh2* mutants prompted us to investigate whether RA exposure could also rescue the absence of pectoral fins in *hoxba;hoxbb* cluster mutants. To explore this, we introduced a frameshift mutation in *raldh2* using CRISPR-Cas9, confirming the phenotypic recapitulation of previously described *raldh2*^{−/−} mutants (figure-supplement 1). Consistent with prior results (Gibert et al., 2006 [↗](#); Grandel et al., 2002 [↗](#)), RA treatment in our *raldh2* mutants rescued the expression of *tbx5a* in pectoral fins (**Figure 4A–D** [↗](#)). In contrast, RA did not induce *tbx5a* expression in the fin buds of *hoxba;hoxbb* cluster mutants (**Figure 4E–H** [↗](#)), and no trace of pectoral fin formation was observed in the treated embryos even at 5 dpf. Furthermore, the endogenous expression patterns of *raldh2* in *hoxba;hoxbb* cluster mutants seem indistinguishable from those in wild-type embryos (**Figure 4I, J** [↗](#)), suggesting that the response to RA is absent in the fin buds of *hoxba;hoxbb* cluster mutants, while RA synthesis appears intact. Although a model proposing that RA directly induces *tbx5a* expression has been suggested (Gibert et al., 2006 [↗](#); Grandel and Brand, 2011 [↗](#); Neto et al., 2012 [↗](#)), our results indicate that RA-dependent induction of *tbx5a* does not occur in the absence of *hoxba* and *hoxbb* clusters. Given that *hoxba;hoxbb* cluster mutants retain the other five intact hox clusters, our findings further support the notion that the *hoxba* and *hoxbb* clusters are specifically responsible for the RA-mediated specification of pectoral fins.

***hoxb5a/b5b* expression in fin buds is RA-dependent**

The absence of pectoral fins, commonly observed in *hoxba;hoxbb* cluster mutants and *raldh2* mutants, suggests that candidate hox genes may be downregulated in *raldh2* mutants. To investigate this, we performed RNA-seq analysis to compare the expression profiles between sibling and *raldh2* homozygous embryos. As expected, we found that most of the 3'-located hox genes are downregulated in *raldh2*^{−/−} embryos (**Figure 5A, B** [↗](#), figure-supplement 2). Although *hoxc1a* showed the most pronounced reduction, our analysis focused on *hoxba* and *hoxbb* clusters. Notably, the levels of *hoxb5b* transcripts were significantly reduced, along with those of *hoxb5a* among *hoxba* genes, in *raldh2* mutants. Previous studies have shown that zebrafish *hoxb5a* and *hoxb5b* are expressed in the lateral plate mesoderm corresponding to presumptive pectoral fin buds (Waxman et al., 2008 [↗](#)). Additionally, knockout mice for *Hoxb5*, which is homologous to

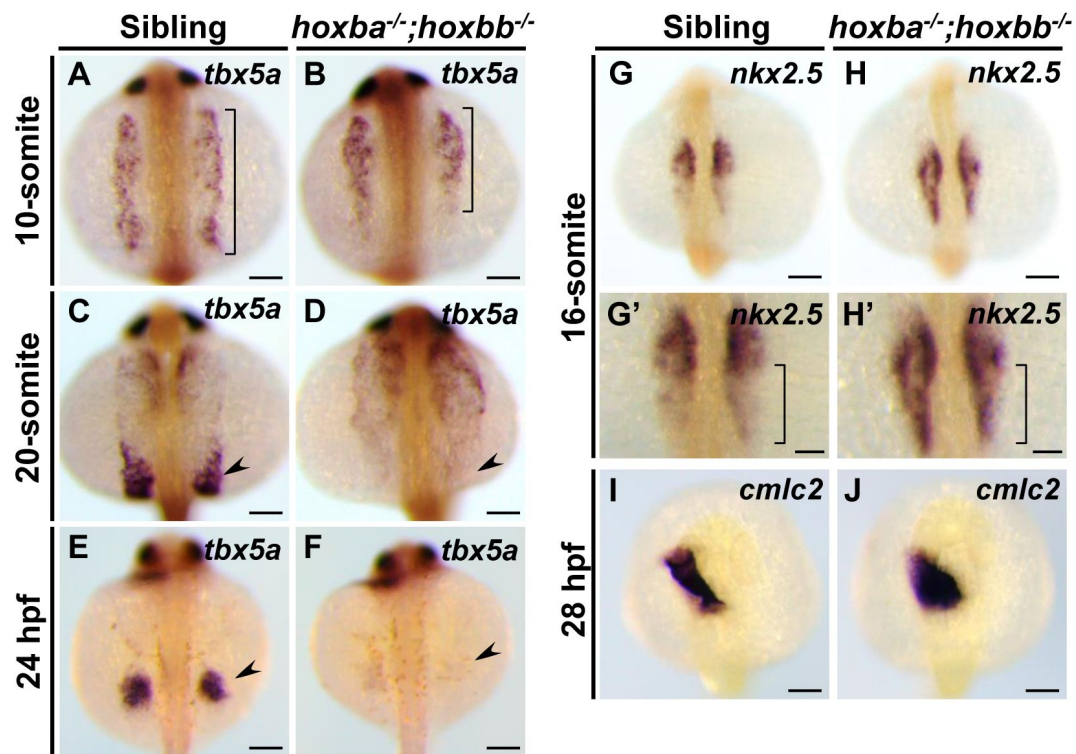


Figure 3.

Loss of fin progenitors with posterior expansion of cardiac marker expression in *hoxba*;*hoxbb* mutants.

(A-F) Expression patterns of *tbx5a* were compared between sibling wild-type and *hoxba*;*hoxbb* homozygous embryos during embryogenesis. The range of *tbx5a* expression in the lateral mesoderm is indicated by a bracket. The progenitor cells of pectoral fins are indicated by an arrowhead. (G, H) Expression patterns of *nkx2.5* were compared between sibling and *hoxba*;*hoxbb* mutants. Enlarged views are shown in (G', H'). The different regions of *nkx2.5* expression between wild-type and mutants are indicated by brackets. (I, J) Expression patterns of *cmlc2* are shown. All images were captured from the dorsal side. For each stage, reproducibility was confirmed with at least three different specimens. Scale bars indicate 100 μ m.

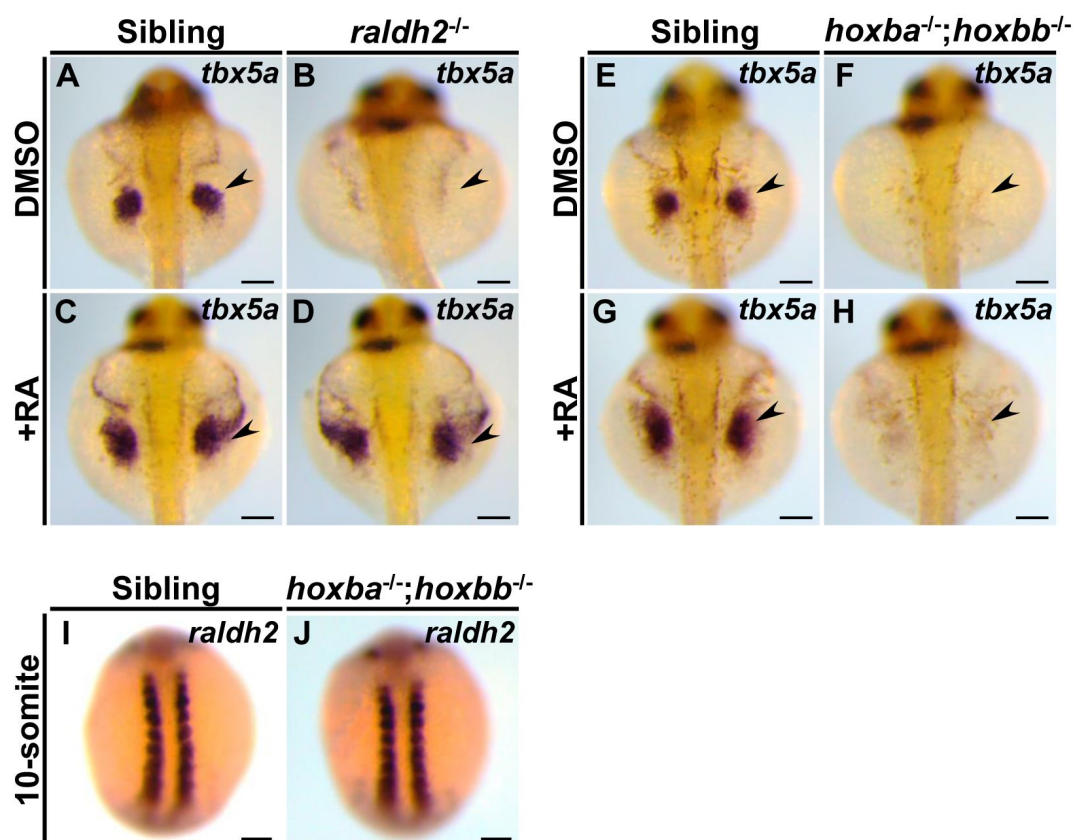


Figure 4.

Zebrafish *hoxba*;*hoxbb* mutants lack a response to RA in fin buds.

(A-D) Exogenous RA exposure in *raldh2* mutants can rescue *tbx5a* expression in fin buds. Arrowheads indicate the positions of the pectoral fin buds. (E-H) RA treatments in *hoxba*;*hoxbb* cluster-deleted mutants do not rescue *tbx5a* expression in fin buds. (I, J) Expression patterns of *raldh2* were compared between sibling and *hoxba*;*hoxbb* mutants. For each genotype, reproducibility was confirmed with at least three different specimens. All images were captured from the dorsal side. Scale bars represent 100 μ m.

zebrafish *hoxb5a* and *hoxb5b*, only exhibit altered positioning of forelimbs (Rancourt et al., 1995 [\[1\]](#)), among various other Hox knockouts that have been generated. Therefore, we examined whether RA regulates the expression of *hoxb5a* and *hoxb5b* in fin buds. In wild-type embryos, *hoxb5a* expression was observed in the neural tube, somites, and lateral plate mesoderm (**Figure 5C** [\[1\]](#)). However, in *raldh2* mutants, *hoxb5a* expression was significantly reduced, notably in the lateral plate mesoderm, which differentiates into the fin buds, where it was reduced to the point of being undetectable (**Figure 5D** [\[1\]](#)). Since RA treatments in *raldh2* mutants can rescue pectoral fin development, we found that *hoxb5a* expression in *raldh2* mutants is also rescued by RA exposure (**Figure 5C-E** [\[1\]](#)). Regarding the expression of *hoxb5b*, which diverged from the same ancestral gene as *hoxb5a*, RA-dependent expression was also confirmed, mirroring the observations made with *hoxb5a* (**Figure 5F-H** [\[1\]](#)). These results suggest that both *hoxb5a* and *hoxb5b* are involved in the formation of the pectoral fins in zebrafish.

Genetic screen for hox genes specifying pectoral fin buds

To understand the molecular mechanisms underlying pectoral fin specification, we sought to identify hox genes essential for fin formation through a genetic approach. We anticipated that frameshift mutations in relevant hox genes would phenocopy the absence of pectoral fins observed in *hoxba*;*hoxbb* mutants. Since *hoxbb* cluster contains fewer hox genes than *hoxba* cluster (**Figure 5A** [\[1\]](#)), we focused our investigation on *hoxbb* cluster. Although *hoxb5b* is expressed in the presumptive fin buds (**Figure 5F** [\[1\]](#)), other *hoxbb* genes may also be involved. Therefore, we simultaneously injected three gRNAs targeting *hoxb5b*, *hoxb6b*, and *hoxb8b* into *hoxba* hemizygous mutants, expecting various germline mutations in these *hoxbb* genes in the injected fish (**Figure 6A** [\[1\]](#)). Subsequently, F1 fish with a hemizygous deletion of *hoxba* cluster and mosaic mutations were crossed with *hoxba*;*hoxbb* hemizygous mutants. Among the progeny, several embryos lacking pectoral fins were identified. Following genotyping and DNA sequencing of the target *hoxbb* genes, we found that all embryos without pectoral fins possessed frameshift mutations in *hoxb5b*, with *hoxba*^{-/-};*hoxb5b*^{+/-} being common among them (**Figure 6B-E** [\[1\]](#), table-supplement 1, figure-supplement 3). Similarly, we injected each crRNA for *hoxb6b* and *hoxb8b* into *hoxba*^{+/-} mutants and confirmed that neither frameshift mutation in *hoxb6b* nor *hoxb8b* in *hoxba*^{-/-};*hoxbb*^{+/-} led to the loss of pectoral fins (**Figure 6F, G** [\[1\]](#), figure-supplement 3). These results suggest that the frameshift mutation of *hoxb5b* alone can phenocopy the deletion of *hoxbb* cluster, recapitulating the absence of pectoral fins in *hoxba*;*hoxbb* cluster mutants.

However, we encountered an unexpected result: *hoxba*^{-/-};*hoxb5b*^{-/-} embryos, which have a homozygous frameshift mutation in *hoxb5b* and a homozygous deletion of *hoxba* cluster, exhibited severely truncated but visibly present pectoral fins (**Figure 6H, I** [\[1\]](#)). After analyzing hundreds of embryos from several crossings, we did not find any *hoxba*^{-/-};*hoxb5b*^{-/-} embryos without pectoral fins. Consistent with this, the expression of *tbx5a* was detectable in the fin buds of *hoxba*^{-/-};*hoxb5b*^{-/-} embryos (**Figure 6K, L** [\[1\]](#)), in contrast to its absence in *hoxba*;*hoxbb* cluster mutants (**Figure 1L** [\[1\]](#)). To examine the potential contributions of other *hoxbb* genes, we observed the phenotypes of *hoxba*^{-/-};*hoxb5b*^{-/-}; *hoxb6b*^{-/-} embryos; however, these mutants did not exacerbate the abnormalities and still formed pectoral fins (**Figure 6J, M** [\[1\]](#)).

The phenotype of *hoxba*^{-/-};*hoxb5b*^{-/-} embryos indicates that a frameshift mutation in *hoxb5b* does not fully mimic the complete deletion of the entire *hoxbb* cluster, suggesting that other mechanisms may be involved. A novel genetic compensation mechanism known as transcriptional adaptation has been identified: mRNA containing a premature termination codon (PTC) can induce increased expression of structurally related genes, thereby compensating for the function of mutated genes (El-Brolosy et al., 2019 [\[2\]](#); Rossi et al., 2015 [\[3\]](#)). To avoid producing transcripts with PTC (Sztal and Stainier, 2020 [\[4\]](#)), we generated a full locus-deleted allele of *hoxb5b* using CRISPR-Cas9 with two crRNAs targeting both ends of the target locus (figure-supplement 4). In contrast to the phenotype of *hoxba*^{-/-};*hoxb5b*^{-/-}, we found that *hoxba*^{-/-};*hoxb5b*^{de/dell} mutants lack pectoral fins and exhibit a significant reduction in *tbx5a* expression when intercrossing

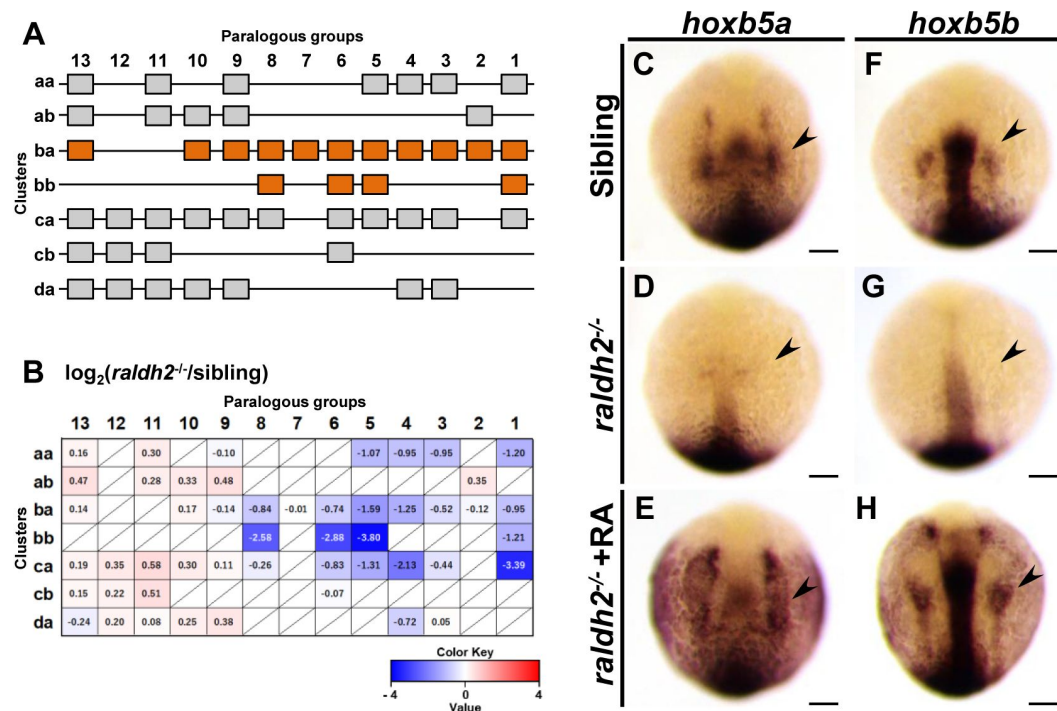


Figure 5.

Expression patterns of *hox5a* and *hoxb5b* are regulated by RA.

(A) Schematic representation of the 49 *hox* genes organized into seven *hox* clusters in zebrafish. *hox* genes in *hoxba* and *hoxbb* clusters are highlighted with orange. (B) Expression profiles of the 49 *hox* genes in wild-type and *raldh2*^{-/-} embryos at the 20-somite stage, analyzed by RNA-seq. The average FPKM of each *hox* gene was compared between sibling and *raldh2* mutants. The absence of specific *hox* genes is indicated by a slash. (C-H) Expression patterns of *hoxb5a* and *hoxb5b* in sibling wild-type, *raldh2*^{-/-}, and RA-treated *raldh2*^{-/-} embryos at the 10-somite stage. Arrowheads indicate the presumptive positions of pectoral fin buds. For each genotype, reproducibility was confirmed with at least three different specimens. All images are captured from the dorsal side. Scale bars represent 100 μ m.

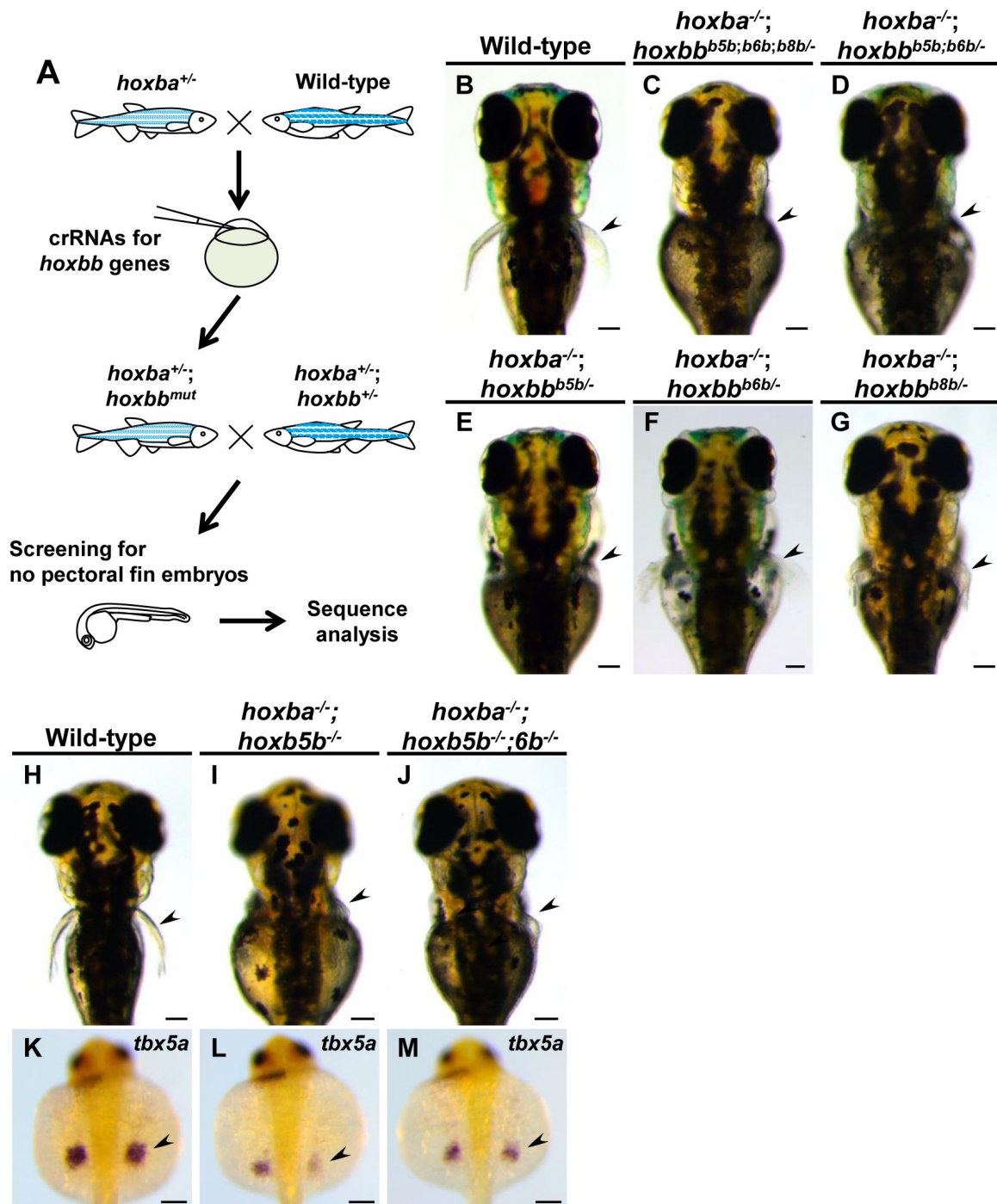


Figure 6.

Screening for hox genes responsible for the zebrafish pectoral fin formation.

(A) Schematic representation of the genetic screening for *hox* genes involved in the specification of pectoral fins in zebrafish. (B-G) Dorsal views of live mutant larvae at 3 dpf, obtained during the screening. After capturing images, genotyping and DNA sequencing in target *hoxb* genes were conducted. For the mutants illustrated in (C-G), *hoxb* genes are shown with frameshift mutations introduced. Detailed information is provided in Supplementary Table S1. (H-J) Dorsal views of *hoxba*^{-/-}; *hoxb5b*^{-/-} and *hoxba*^{-/-}; *hoxb5b*^{-/-}; *hoxb6b*^{-/-} larvae at 3 dpf. (K-M) Expression patterns of *tbx5a* in *hoxba*^{-/-}; *hoxb5b*^{-/-} and *hoxba*^{-/-}; *hoxb5b*^{-/-}; *hoxb6b*^{-/-} mutants at 30 hpf. Arrowheads indicate the presumptive positions of pectoral fin buds. All images are captured from the dorsal side. Scale bars represent 100 μ m.

hoxba^{+/-};*hoxb5b*^{+/-del} fish (**Figure 7A, B, D, E**). However, the occurrence rate of embryos lacking pectoral fins ($n=3/120$; 2.5 %) was lower than predicted based on Mendelian genetics (1/16; 6.3 %). Other *hoxba*^{+/-};*hoxb5b*^{del/del} embryos did exhibit shortened pectoral fins (**Figure 7C**). Additionally, concerning *hoxba* cluster, we also generated frameshift-induced *hoxb5a*^{+/-};*hoxb5b*^{+/-} and a full locus-deleted *hoxb5a*^{del/del};*hoxb5b*^{del/del} embryos (figure-supplement 4). *hoxb5a*^{+/-};*hoxb5b*^{+/-} larvae consistently displayed normal pectoral fins (**Figure 7F**). In contrast, *hoxb5a*^{del/del};*hoxb5b*^{del/del} mutants exhibited a slightly more pronounced phenotype of shortened pectoral fins, but no instances of missing pectoral fins were identified (**Figure 7G**). To further clarify which Hox genes could potentially contribute to fin bud formation, we analyzed the expression of PG4 and PG6–8 genes from *hoxba* and *hoxbb* clusters using whole-mount *in situ* hybridization. Among these, only *hoxb4a* showed detectable expression in the fin buds, whereas *hoxb6a*, *hoxb6b*, *hoxb7a*, *hoxb8a*, and *hoxb8b* were not detectable (**Figure 7M–R**). Furthermore, due to the functional redundancy observed between *hoxb4a* and *hoxb5a* observed in zebrafish vertebral patterning and the RA-dependent expression of *hoxb4a* (Grandel et al., 2002; Maeno et al., 2024), we generated frameshift-induced *hoxb4a*^{+/-};*hoxb5a*^{+/-};*hoxb5b*^{+/-} embryos and an allele lacking the entire genomic region encompassing *hoxb4a* and *hoxb5a* (figure-supplement 4). Although *hoxb4a*^{+/-};*hoxb5a*^{+/-};*hoxb5b*^{+/-} embryos did not lack pectoral fins (**Figure 7H**), some *hoxb4a-b5a*^{del/del};*hoxb5b*^{del/del} embryos did lack pectoral fins, although at low penetrance (**Figure 7I**, $n=3/397$; 0.7%). The majority of *hoxb4a-b5a*^{del/del};*hoxb5b*^{del/del} embryos displayed truncated pectoral fins (**Figure 7J**). The lack of pectoral fins was not detected in *hoxb4a-b5a*^{del/del} and *hoxb4a*^{del/del};*hoxb5b*^{del/del} mutants (**Figure 7K, L**, figure-supplement 4). These results suggest that *hoxb4a*, *hoxb5a*, and *hoxb5b* may cooperatively contribute to the anterior-posterior positioning of pectoral fin buds in zebrafish. Furthermore, our findings reveal that transcriptional adaptation may partially play a role in genetic compensation for frameshift-induced *hox* mutations during pectoral fin formation, while also suggesting the potential involvement of other unknown mechanisms or additional genomic regions in the initial induction of zebrafish pectoral fin buds.

Discussion

Since the establishment of gene targeting (Mansour et al., 1988), numerous Hox knockout mice have been generated. However, neither single nor compound Hox mutants, nor Hox overexpression approaches, have been reported to cause significant defects in limb positioning (Jurberg et al., 2013; Moreau et al., 2019). Consequently, the role of Hox genes in the positioning of vertebrate paired appendages remains unclear due to a lack of direct genetic evidence. In this study, we provide genetic evidence using zebrafish, demonstrating that specific combinatorial deletions of *hoxba* and *hoxbb* clusters result in the absence of pectoral fins and a lack of induction of *tbx5a* expression in the pectoral fin field. Based on our results, we propose a model in which Hox expressions in the lateral plate mesoderm, regulated by RA synthesized in the paraxial mesoderm, provide positional information and induce *tbx5a* expression in the presumptive fin buds. Our genetic results shed light on longstanding fundamental questions regarding the molecular mechanisms by which the positioning of paired appendages is established in vertebrates.

hoxba/bb clusters (possibly *hoxb4a*, *b5a*, *b5b*) define the pectoral fin field

We showed that *hoxba*;*hoxba* mutants exhibit a lack of pectoral fins. Abnormal expression of *tbx5a* in the lateral plate mesoderm, where pectoral fin precursor cells normally arise, was evident from its initial expression in the mutants. We presume that *hox* genes in *hoxba* and *hoxbb* clusters, induced by RA, define the region along the anterior-posterior axis where pectoral fin formation can occur and promote the expression of *tbx5a* in the lateral plate mesoderm. Importantly, *tbx5a* expression in the fin bud was not induced even by RA exposure in *hoxba*;*hoxbb* mutants (**Figure**

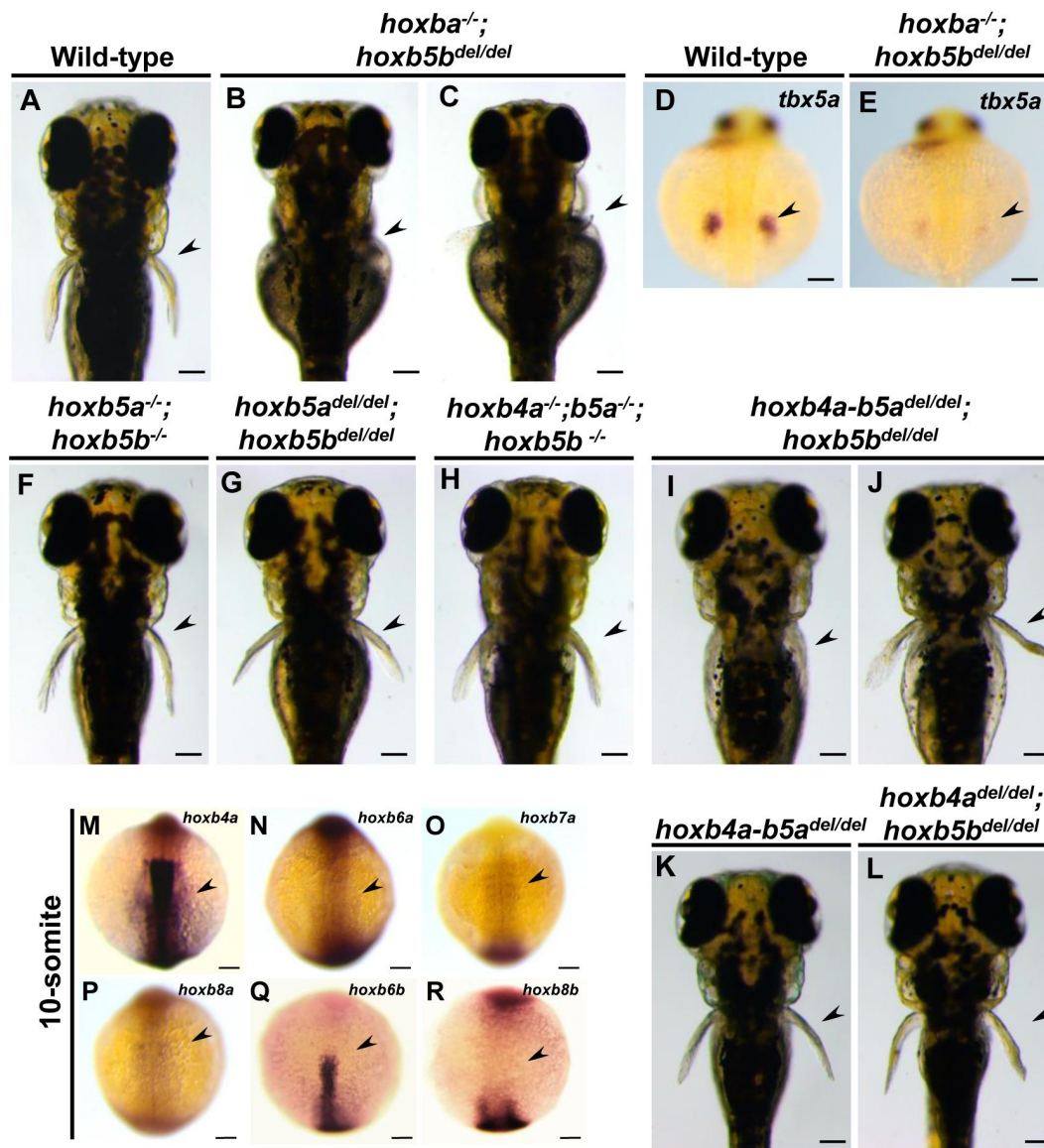


Figure 7.

***hoxb4a-b5a*^{del/del};*hoxb5b*^{del/del} larvae partially recapitulate the absence of the pectoral fins.**

(A-C) Dorsal views of zebrafish *hoxba*^{-/-};*hoxb5b*^{del/del} larvae at 3 dpf. (D, E) Expression patterns of *tbx5a* in sibling wild-type and *hoxba*^{-/-};*hoxb5b*^{del/del} (*n*=4) at 30 hpf. Arrowheads indicate the presumptive positions of pectoral fin buds. (F, G) Dorsal view of frameshift-induced *hoxb5a*^{-/-};*hoxb5b*^{-/-} and *hoxb5a*^{del/del};*hoxb5b*^{del/del} larvae. (I-K) Dorsal view of frameshift-induced *hoxb4a*^{-/-};*hoxb5a*^{-/-};*hoxb5b*^{-/-} and *hoxb4a-b5a*^{del/del};*hoxb5b*^{del/del} larvae. All images are captured from the dorsal side. Arrowheads indicate the presumptive positions of pectoral fin buds. (M-R) Expression patterns of zebrafish *hoxb4a*, *hoxb6a*, *hoxb6b*, *hoxb7a*, *hoxb8a*, and *hoxb8b* at the 10-somite stage. Dorsal views. Arrowheads indicate the presumptive positions of pectoral fin buds. Scale bars represent 100 μm.

4E-H [↗](#)). In zebrafish, it has been suggested that RA synthesized in the paraxial mesoderm acts directly on the lateral plate mesoderm, or that RA induces *tbx5a* expression via *wnt2b* in the intermediate mesoderm (Gibert et al., 2006 [↗](#); Grandel and Brand, 2011 [↗](#); Neto et al., 2012 [↗](#)). However, our results suggest that intact *hoxba* and *hoxbb* clusters are required for RA-dependent induction of *tbx5a* expression. One possible explanation is a linear pathway in which RA induces *hox* expression, which then activates *tbx5a*. Alternatively, RA may induce *hox* expression and, together with Hox proteins, act cooperatively to activate *tbx5a*. Our data are consistent with both models. Through genetic studies, albeit with low penetrance, we further showed that *hoxb4a*, *hoxb5a*, and *hoxb5b* play a significant role in the induction of pectoral fin buds. The homologous Hoxb4 and Hoxb5 proteins in mice have been shown to bind directly to the enhancer of *Tbx5* (Minguillon et al., 2012 [↗](#)). Moreover, in chickens, *Tbx5* expression in the forelimb field is restricted by posterior Hox genes, preventing posterior expansion (Moreau et al., 2019 [↗](#); Nishimoto et al., 2014 [↗](#)). In contrast, our combinatorial deletions of zebrafish *hox* clusters did not produce posterior expansion of *tbx5a* expression. We therefore propose that, in zebrafish, *hoxb4a*, *hoxb5a*, and *hoxb5b*, induced by RA in the paraxial mesoderm, act primarily to promote *tbx5a* expression in the pectoral fin field.

Partial phenocopy of locus-deleted *hox* mutants implies other unidentified mechanisms

In this study, we observed that *hox* mutants with deletions of genomic loci exhibited more pronounced pectoral fin phenotypes than frameshift-induced *hox* mutants. The frameshift mutants used in this study are likely to be loss-of-function alleles, as we demonstrated several defects in vertebral patterning in these mutants (Maeno et al., 2024 [↗](#)). These results suggest that transcriptional adaptation may occur during the formation of pectoral fins in frameshift-induced *hox* mutants, potentially compensating for the loss of function. In our previous studies, which focused on dorsal and anal fin patterning, vertebral patterning, and swim bladder formation, we demonstrated that the phenotypes observed in zebrafish *hox* cluster-deleted mutants could essentially be replicated by introducing frameshift mutations into *hox* genes within the cluster (Adachi et al., 2024 [↗](#); Maeno et al., 2024 [↗](#); Satoh et al., 2024 [↗](#); Yamada et al., 2021 [↗](#)). Therefore, creating a mutant with a deleted locus that does not induce transcriptional adaptation was unnecessary. One factor that may explain the differences in results between this study and our prior findings is that even a small amount of *hox* gene products essential for pectoral fin positioning may be sufficient to induce fin buds, as shown in **Figure 1** [↗](#), potentially leading to the manifestation of gene compensation through transcriptional adaptation in frameshift-induced mutants. Alternatively, transcriptional adaptation may function differently depending on various developmental processes or stages. Interestingly, all the aforementioned phenotypes reproducible by frameshift mutations were observed at later developmental stages than pectoral fin formation. Additionally, in *hox* locus-deleted mutants, only some *hoxba*^{-/-};*hoxb5b*^{de/del} or *hoxb4a-b5a*^{de/del};*hoxb5b*^{de/del} embryos were able to reproduce the loss of the pectoral fin, suggesting the presence of other compensatory mechanisms. The organized expression patterns of Hox genes are thought to be regulated by complex regulatory controls, with their expression regulated according to the genomic order within the cluster. In the absence of Hox function, neighboring intact Hox genes could compensate for the loss of function. It is also possible that unidentified non-coding regions in *hoxba* and *hoxbb* clusters are involved in pectoral fin positioning. These issues should be clarified in future studies.

Hox genes and the evolutionary origin of pectoral fins

According to fossil records from the Cambrian period, the earliest vertebrates did not possess paired fins (Janvier, 1996 [↗](#); Shubin et al., 1997 [↗](#)). Later in evolution, vertebrates acquired primitive pectoral fins, which served as evolutionary precursors to tetrapod forelimbs, before diverging into ray-finned and lobe-finned fishes. Our results align well with this evolutionary scenario. Hox genes, which are conserved across bilaterian animals, were already present in primitive vertebrates without paired fins, suggesting that their functional diversification may have

facilitated the origin of new appendages. The early fossils of vertebrates with pectoral fins were discovered in Ordovician and Silurian jawless fishes (Janvier, 1996 [↗](#); Shubin et al., 1997 [↗](#)). Our finding that *hoxba*;*hoxbb* mutants completely lack pectoral fins implies that molecular evolution within the ancestral HoxB cluster contributed to the emergence of pectoral fins. Although the precise molecular modifications remain unclear, one plausible model is that HoxB genes acquired the capacity to induce *Tbx5* expression in the lateral plate mesoderm, establishing the RA–HoxB–*Tbx5* gene network essential for fin initiation. We presume that the acquisition of pectoral fins was driven by multiple genomic changes accumulated over time, among which molecular evolution within the HoxB cluster may have represented one of the principal factors contributing to the origin of this novel appendage.

The positioning of pectoral fins is restricted to specific hox clusters in zebrafish, in contrast to the functional overlap seen in mice

Taking advantage of our availability of seven individual hox cluster mutants in zebrafish, we generated multiple cluster deletions and demonstrated that hox genes required for establishing pectoral fin buds are restricted to *hoxba* and *hoxbb* clusters. In contrast, analyses of knockout mice have revealed extensive functional overlap among the four Hox clusters. For instance, mice lacking all HoxB genes except for *Hoxb13* do not lose forelimbs (Medina-Martinez et al., 2000 [↗](#)), and forelimbs are still present in *Hoxa5*;*Hoxb5*;*Hoxc5* triple knockout mice, which lack all Hox genes of the paralog group 5 (Xu et al., 2013 [↗](#)). Thus, while zebrafish *hoxba*;*hoxbb* mutants exhibit a striking phenotype of absent pectoral fins, forelimbs are preserved in comparable mouse mutants. We speculate that paired fins emerged after the quadruplication of primitive Hox clusters, and that novel functions related to pectoral fin formation arose within the HoxB cluster. In zebrafish, these functions remain confined to *hoxba* and *hoxbb* clusters, making them indispensable for pectoral fin development. Interestingly, a similar cluster-specific confinement has been observed in median fins: in both zebrafish and medaka, Hox genes responsible for positioning dorsal and anal fins are primarily located in HoxC-related clusters (Adachi et al., 2024 [↗](#)). In contrast, in mice, functional redundancy among Hox clusters—likely accumulated during over half a billion years of vertebrate evolution—may mask the requirement of HoxB genes for forelimb formation. While the precise evolutionary mechanisms remain to be clarified, our results suggest that cluster-specific specialization in teleosts contrasts with broader redundancy across Hox clusters in mammals.

Materials and methods

Zebrafish

Riken Wild-type (RW) zebrafish, obtained from the National BioResource Project in Japan, were maintained at 27°C with a 14-hour light/10-hour dark cycle. Embryos were collected from natural spawning, and the larvae were raised at 28.5°C. Developmental stages of the embryos and larvae were determined based on hours post-fertilization (hpf), days post-fertilization (dpf), and developmental stage-specific features (Kimmel et al., 1995 [↗](#); Parichy et al., 2009 [↗](#)). The alleles of *hox* cluster-deleted mutants used in this study were previously generated using CRISPR-Cas9 (Yamada et al., 2021 [↗](#)). The following frameshift-induced *hox* mutants were created using CRISPR-Cas9: *hoxb5a*^{sud125}, *hoxb5b*^{sud136}, *hoxb6b*^{sud137}, and *hoxb4a*;*hoxb5a*^{sud144} (Maeno et al., 2024 [↗](#)). All experiments involving live zebrafish were approved by the Committee for Animal Care and Use of Saitama University.

Generation of mutants by the CRISPR-Cas9 system

All the mutants used in this study were generated using the Alt-R CRISPR-Cas9 system (Integrated DNA Technologies). To create frameshift-induced mutants, gene-specific crRNAs were incubated with common tracrRNA, followed by Cas9 nuclease. For the generation of mutants that delete large

genomic regions, two crRNAs targeting both ends of the regions were incubated with common tracrRNA and Cas9 nuclease. Approximately one nanoliter of the crRNA:tracrRNA-Cas9 RNA-protein complex was injected into fertilized embryos, which were subsequently raised to juvenile fish. Candidate founder fish were selected using a heteroduplex mobility shift assay for frameshift mutants (Ota et al., 2013 [DOI](#)) and by amplifying genomic deletions with specific primers. After sexual maturation, candidate founder fish were mated with wild-type fish to produce heterozygous F1 offspring. Among the F1 offspring, mutant fish carrying the same mutation were identified through PCR-based genotyping followed by DNA sequencing. The target-specific sequences of the crRNAs used in this study are listed (table-supplement 2). The frozen sperm from the mutants used in this study have been deposited in the National BioResource Project Zebrafish in Japan (<https://shigen.nig.ac.jp/zebra/> [DOI](#)) and are available upon request.

Genotyping of mutants

For the phenotypic analysis of embryos and the maintenance of mutant fish lines, PCR-based genotyping was performed. Briefly, genomic DNA was extracted from the analyzed embryos or the partially dissected caudal fins of anesthetized larvae or fish using the NaOH method (Meeker et al., 2007 [DOI](#)), which served as the template for PCR. The sequences of the primers used for genotyping frameshift-induced mutants are listed in the supplementary Table 3. For genotyping deletion mutants of *hox* genes, PCR was conducted using a combination of three primers, with their sequences also provided (table-supplement 3). Genotyping of *hox* cluster-deleted mutants was performed using PCR as previously described (Yamada et al., 2021 [DOI](#)). After the reactions, the PCR products were separated by electrophoresis. Based on differences in the lengths of PCR products derived from wild-type and mutated alleles, either a 2 % agarose gel in 0.5x TBE buffer, a 15 % polyacrylamide gel in 0.5x TBE buffer, or direct DNA sequencing was utilized to determine the genotype.

Whole-mount *in situ* hybridization

Whole-mount *in situ* hybridization was carried out as described (Thisse and Thisse, 2014 [DOI](#)). After the staining, the embryos mounted in 70–80 % glycerol were captured under the stereomicroscope (Leica M205 FA) with a digital camera (Leica DFC350F). After taking images, PCR-based genotyping was carried out as described above.

RNA-seq analysis

For RNA-seq analysis, embryos were obtained by the intercrosses between *raldh2* heterozygous fish. At the 18–20-somite stages, sibling and homozygous embryos were separated based on the phenotype, and 20 embryos per tube were collected. Then, total RNA was extracted by ISOGENE (Nippon Gene) and followed by DNase I treatment (Takara). After the quality check of the isolated RNA, RNA-seq analyses ($n=2$ for each) were carried out by Genewiz, Azenta Life Science.

Treatment of retinoic acid in the embryos

Treatment of retinoic acid on the embryos was performed as previously described, with minor modifications (Gibert et al., 2006 [DOI](#); Grandel et al., 2002 [DOI](#)). Briefly, embryos were obtained from intercrosses between *raldh2* heterozygous mutants. At the 70 % epiboly stage, the embryos were soaked in a solution containing DMSO or 10^{-8} M RA and incubated at 28.5°C. Embryos were fixed at 30 hpf for analysis of *tbx5a* expression and at the 10-somite stage for examination of *hox* gene expression.

Data availability

All data necessary to support the conclusions of this article are provided within the main text, figures, and supplementary data. The other data sets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Additional files

Supplementary Figure and Table [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors have used gene deletion approaches in zebrafish to investigate the function of genes of the hox clusters in pectoral fin "positioning" (but perhaps more accurately pectoral fin "formation")

Strengths:

The authors have employed a robust and extensive genetic approach to tackle an important and unresolved question.

The results are largely very clearly presented.

Weaknesses:

The Abstract suggests that no genetic evidence exists in model organisms for a role of Hox genes in limb positioning. There are, however, several examples in mouse and other models (both classical genetic and other) providing evidence for a role of Hox genes in limb position, which is elaborated on in the Introduction.

It would perhaps be more accurate to state that several lines of evidence in a range of model organisms (including the mouse) support a role for Hox genes in limb positioning. The author's work is not weakened by a more inclusive introduction that cites the current literature more comprehensively.

It would be helpful for the authors to make a clear distinction between "positioning" of the limb/fin and whether a limb/fin "forms" at all, independent of the relative position of this event along the body axis.

Discussion of why the zebrafish is sensitive to Hoxb loss with reference to the fin, but mouse Hoxb mutants do make a limb?

Is this down to exclusive expression of Hoxbs in the zebrafish pectoral fin forming region rather than a specific functional role of the protein? This is important as it has implications for the interpretation of results throughout the paper and could explain some apparently conflicting results. .

Why is hoxba more potent than hoxbb? Is this because Hoxba has Hox4/5 present while hoxbb has only hoxb5? Hoxba locus has retained many more hox genes in, cluater than hoxbb therefore might expect to see greater redundancy in this locus)

Deletion of either hox a or hox d in background of hoxba mutant does have some effect. IS this a reflection of protein function or expression dynamics of hoax/hoxd genes?

Can we really be confident there is a "transformation of pectoral fin progenitor cells into cardiac cells"?

The failure to repress Nkx2.5 in the posterior (pelvic fin) domain is clear but have these cells actually acquired cardiac identity? They would be expected to express Tbx5a (or b) as cardiac precursors but this domain does not broaden. There is no apparent expansion of the heart (field)/domain or progenitors beyond 16 somite stage. The claimed "migration" of heart precursors iin the mutant is not clear. The heart/cardiac domain that does form in the mutant is not clearly expanded in the mutant. The domain of cmlc2 looks abnormal in the mutant but I am not convinced it is "enlarged" as claim by the authors. The authors have not convincingly shown that " the cells that should form the pectoral fin instead differentiate into cardia cells."

The only clear conclusion is the loss of pectoral fin-forming cells rather than these fin-forming cells being "transformed" into a new identity. It would be interesting to know what has happened to the cells of the pectoral fin forming region in these double mutants.

It is not clear what the authors mean by a "converse" relationship between forelimb/pectoral fin and heart formation. The embryological relationship between these two populations is distinct in amniotes.

The authors show convincing data that RA cannot induce Tbx5a in the absence of Hob clusters but I am not convinced by the interpretation of this result. The results shown would still be consistent with RA acting directly upstream of tbx5a but merely that RA acts in concert with hox genes to activate tbx5a. IN the absence of one or the other tbx5a would not be expressed. It is not necessary that RA and hoxbs act exclusively in a linear manner (i.e. RA regulates hoxb that in turn regulate tbx5a)

The authors have carried out a functional test for the function of hoxb6 and hoxb8 in the hemizygous hoxb mutant background. What is lacking is any expression analysis to demonstrate whether hoxb6b or hoxb8b are even expressed in the appropriate pectoral fin territory to be able to contribute to pectoral fin development either in this assay or in normal pectoral fin development.

(The term "compensate" used in this section is confusing/misleading.)

The authors' confounding results described in Figures 6-7 are consistent with the challenges faced in other model organisms in trying to explore the function of genes in the hox cluster

and the known redundancy that exists across paralogous groups and across individual clusters.

Given the experimental challenges in deciphering the actual functions of individual or groups of hox genes, a discussion of the normal expression pattern of individual and groups of hox genes (and how this may change in different mutant backgrounds) could be helpful to make conclusions about likely normal function of these genes and compensation/redundancy in different mutant scenarios.

Comments on revisions:

No further issues to address.

<https://doi.org/10.7554/eLife.105889.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors of this manuscript performed a fascinating set of zebrafish mutant analysis on hox cluster deletion and pinpoint the cause of the pectoral fin loss in one combinatorial hox cluster mutant of *hoxba* and *hoxbb*. I support the publication of this manuscript.

Strengths:

The study is based on a variety of existing experimental tools that enabled the authors' past construction of hox cluster mutants and is well-designed. The manuscript is well written to report the author's findings on the mechanism that positions the pectoral fin.

Weaknesses:

The study does not focus on the other hox clusters than *ba* and *bb*, and is confined to the use of zebrafish, as well as the comparison with existing reports from mouse experiments.

Comments on revisions:

The authors have sufficiently addressed the concerns raised in my previous review. The revised manuscript substantially strengthens the original work.

<https://doi.org/10.7554/eLife.105889.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors have used gene deletion approaches in zebrafish to investigate the function of genes of the hox clusters in pectoral fin "positioning" (but perhaps more accurately pectoral fin "formation").

Strengths:

The authors have employed a robust and extensive genetic approach to tackle an important and unresolved question. The results are largely presented in a very clear way.

We thank the reviewer for the positive summary and for recognizing the strengths of our genetic approach and presentation.

Weaknesses:

The Abstract suggests that no genetic evidence exists in model organisms for a role of Hox genes in limb positioning. There are, however, several examples in mouse and other models (both classical genetic and other) providing evidence for a role of Hox genes in limb position, which is elaborated on in the Introduction.

It would perhaps be more accurate to state that several lines of evidence in a range of model organisms (including the mouse) support a role for Hox genes in limb positioning. The author's work is not weakened by a more inclusive introduction that cites the current literature more comprehensively.

Thank you for this constructive comment. We agree that our Abstract implied an absence of genetic evidence across model organisms and could be misleading. We have revised the Abstract to acknowledge that multiple lines of evidence—including classical and molecular studies in mouse and other models—support a role for Hox genes in limb/fin positioning. We have also expanded the Introduction to cite this literature more comprehensively. These changes clarify the current state of knowledge while preserving the novelty of our zebrafish genetic findings.

It would be helpful for the authors to make a clear distinction between "positioning" of the limb/fin and whether a limb/fin "forms" at all, independent of the relative position of this event along the body axis.

We thank the reviewer for pointing this out. In the revised manuscript, we now make a distinction between these two aspects: we describe “positioning” as being specified by the expression domains of Hox genes along the anterior–posterior axis, while the “formation” of pectoral fins reflects the functional requirement of Hox genes to induce *tbx5a* expression and thereby initiate fin development. We have clarified this distinction in the text to better separate these related but distinct roles of Hox genes.

Discussion of why the zebrafish is sensitive to Hoxb loss with reference to the fin, but mouse Hoxb mutants do make a limb?

We thank the reviewer for this important comment. Our interpretation is that paired fins first appeared in vertebrates that already possessed four Hox clusters. It is likely that novel functions related to pectoral fin positioning emerged within the HoxB cluster at that time, contributing to the origin of pectoral fins. In zebrafish, we found that these functions remain largely restricted to the *hoxba* and *hoxbb* clusters, such that loss of both results in complete absence of pectoral fins. In contrast, mice exhibit a high degree of functional redundancy across Hox clusters. For example, deletion of all HoxB genes except *Hoxb13* does not result in forelimb loss (Medina-Martinez et al., 2000), and forelimbs are still present in *Hoxa5;Hoxb5;Hoxc5* triple knockouts (Xu et al., 2013). Thus, although we cannot fully explain why HoxB cluster deletions alone do not abolish forelimb formation in mice, it is plausible that overlapping functions from other Hox clusters compensate for the loss of HoxB genes, consistent with the general robustness of the mammalian Hox system. We have revised the Discussion to clarify this point.

Is this down to exclusive expression of Hoxbs in the zebrafish pectoral fin forming region rather than a specific functional role of the protein? This is important as it has

implications for the interpretation of results throughout the paper and could explain some apparently conflicting results.

We thank the reviewer for this insightful comment. To address this point, we newly analyzed the expression patterns of PG4–8 genes in the *hoxba* and *hoxbb* clusters. Our in situ hybridization results revealed that only *hoxb4a*, *hoxb5a*, and *hoxb5b* are detectably expressed in the pectoral fin buds (Figure 5C, 5E, Figure 7M–R). While we cannot completely exclude the possibility of functional differences among Hox proteins, our data strongly suggest that the loss of pectoral fins in *hoxba*;*hoxbb* cluster mutants is primarily due to the expression domains of these specific Hox genes in the fin-forming region, rather than to unique biochemical properties of the proteins. We have added these new data as a figure in the revised manuscript (Figure 7M–R) and clarified this point in the text (lines 312–316).

Why is Hoxba more potent than Hoxbb? Is this because Hoxba has Hox4/5 present, while Hoxbb has only Hoxb5? Hoxba locus has retained many more Hox genes in cluster than hoxbb; therefore, one might expect to see greater redundancy in this locus).

We thank the reviewer for raising this important point. At present, we do not know the precise reason why *hoxba* appears more potent than *hoxbb*. The possibility raised by the reviewer—that differences in retained gene content (e.g., Hox4/5 in *hoxba* versus only Hoxb5 in *hoxbb*) may underlie this discrepancy—is certainly plausible. However, our previous study on the formation of dorsal and anal fins showed a similar situation: although PG11–13 Hox genes are present in both *hoxca* and *hoxcb* clusters, deletion of *hox* genes in *hoxca* cluster had a more pronounced effect on median fin development (Adachi et al., 2024). This suggests that, following the teleost-specific whole-genome duplication, duplicated Hox clusters are not functionally equivalent, and asymmetric retention or deployment of functions may occur. The mechanistic basis of such bias remains unclear and warrants further investigation.

Deletion of either Hoxa or Hoxd in the background of the Hoxba mutant does have some effect. Is this a reflection of protein function or expression dynamics of Hoxa/Hoxd genes?

We appreciate the reviewer's comment and the opportunity to clarify this point. In Figure 2, we compared several double mutants with the *hoxba* single mutant. Among thesm, only the *hoxba*;*hoxbb* mutant exhibited a complete loss of *tbx5a* expression, whereas other combinations did not differ substantially from the *hoxba* mutant alone. Therefore, we consider that additional deletions such as *hoxaa*, *hoxab*, and *hoxda* do not have a strong effect beyond the *hoxba* deletion itself, and it is unlikely that Hoxa or Hoxd proteins functionally compensate for Hoxba in regulating *tbx5a* expression. Consistent with this interpretation, in our previous study we did not detect abnormalities in *tbx5a* expression in the *hoxaa*;*hoxab*;*hoxda* triple mutant (Ishizaka et al., 2024). Taken together, these observations support our view that the *hoxba* and *hoxbb* clusters are specifically required for the induction of *tbx5a* in the pectoral fin field.

Can we really be confident that there is a "transformation of pectoral fin progenitor cells into cardiac cells"?

*The failure to repress Nkx2.5 in the posterior (pelvic fin) domain is clear, but have these cells actually acquired cardiac identity? They would be expected to express Tbx5a (or b) as cardiac precursors, but this domain does not broaden. There is no apparent expansion of the heart (field)/domain or progenitors beyond the 16 somite stage. The claimed "migration" of heart precursors in the mutant is not clear. The heart/cardiac domain that does form in the mutant is not clearly expanded in the mutant. The domain of *cmlc2* looks abnormal in the mutant, but I am not convinced it is "enlarged" as*

claimed by the authors. The authors have not convincingly shown that "the cells that should form the pectoral fin instead differentiate into cardiac cells." The only clear conclusion is the loss of pectoral fin-forming cells rather than these fin-forming cells being "transformed" into a new identity. It would be interesting to know what has happened to the cells of the pectoral fin-forming region in these double mutants.

We sincerely thank the reviewer for this important comment. We agree that our data do not yet allow us to conclude with certainty that the presumptive pectoral fin progenitor cells in *hoxba;hoxbb* cluster mutants are fully “transformed” into cardiac cells. Our intention was to describe the striking posterior expansion of *nkx2.5* expression and the altered morphology of the *cmlc2*-positive cardiac field in the mutants, which suggested a shift in cell fate. However, as the reviewer correctly points out, we did not directly demonstrate that the missing fin progenitors acquire bona fide cardiac identity.

To address this, we have revised the text to clarify that the most robust conclusion from our current dataset is the loss of pectoral fin-forming cells in *hoxba;hoxbb* cluster mutants. We have softened or removed the claim of “transformation” and instead emphasize that our observations are consistent with an expansion of cardiac marker expression domains into the region where fin progenitors normally arise. We also acknowledge that the *cmlc2* domain is abnormal rather than unequivocally enlarged, and have adjusted our wording accordingly.

It is not clear what the authors mean by a "converse" relationship between forelimb/pectoral fin and heart formation. The embryological relationship between these two populations is distinct in amniotes.

We thank the reviewer for pointing this out. Our intention was to highlight the reciprocal balance between pectoral fin and cardiac progenitors in zebrafish. In particular, Waxman et al. (2008) demonstrated that retinoic acid signaling promotes pectoral fin formation while restricting the expansion of cardiac progenitors, thereby illustrating this reciprocal relationship. To avoid confusion, we have revised the text to explicitly state that this applies to zebrafish.

*The authors show convincing data that RA cannot induce *Tbx5a* in the absence of *Hob* clusters, but I am not convinced by the interpretation of this result. The results shown would still be consistent with RA acting directly upstream of *tbx5a*, but merely that RA acts in concert with *hox* genes to activate *tbx5a*. In the absence of one or the other, *Tbx5a* would not be expressed. It is not necessary that RA and *hoxbs* act exclusively in a linear manner (i.e., RA regulates *hoxb* that in turn regulates *tbx5a*).*

We appreciate the reviewer’s thoughtful comment. We agree that our original wording in the Results section implied a strictly linear model of RA → Hox → *tbx5a*. In response, we have revised the Results to state only the experimental observation, namely that RA-dependent induction of *tbx5a* does not occur in the absence of the *hoxba* and *hoxbb* clusters.

We have moved the broader interpretation to the Discussion, where we now emphasize that our data are compatible with multiple models. One possibility is a linear pathway in which RA induces Hox expression that subsequently activates *tbx5a*. Alternatively, it is also plausible that RA induces Hox expression and that RA and Hox proteins act cooperatively to induce *tbx5a*. Our findings do not distinguish between these possibilities, and both models remain consistent with the data. We believe this restructuring addresses the reviewer’s concern by keeping the Results factual and limiting mechanistic interpretation to the Discussion.

*The authors have carried out a functional test for the function of *hoxb6* and *hoxb8* in the hemizygous *hoxb* mutant background. What is lacking is any expression analysis to*

demonstrate whether Hoxb6b or Hoxb8b are even expressed in the appropriate pectoral fin territory to be able to contribute to pectoral fin development, either in this assay or in normal pectoral fin development.

We thank the reviewer for emphasizing the importance of expression analyses. In response, we performed a comprehensive whole-mount in situ hybridization survey of all eight PG4–8 Hox genes from the hoxba and hoxbb clusters (hoxb4a, hoxb5a, hoxb5b, hoxb6a, hoxb6b, hoxb7a, hoxb8a, and hoxb8b) during pectoral fin development (18–30 hpf). Among these, only hoxb4a, hoxb5a, and hoxb5b displayed detectable expression in the developing pectoral fin buds. In contrast, hoxb6a, hoxb6b, hoxb7a, hoxb8a, and hoxb8b were not expressed in this territory. These new data have been incorporated into the revised manuscript (Fig. 7M–R). We believe that this dataset provides a more complete and systematic picture of which Hoxb genes are available to function in pectoral fin development, and we are grateful to the reviewer for this valuable suggestion, which significantly strengthened our study.

(The term "compensate" used in this section is confusing/misleading.)

We thank the reviewer for this helpful remark. We agree that the term “compensate” was misleading in this context, as it could be confused with genetic compensation mechanisms such as transcriptional adaptation. To avoid this ambiguity, we have revised the wording.

Specifically, we replaced “compensate for” with “mimic the effect of” or “phenocopy” depending on the context. We believe this revision improves clarity and prevents misunderstanding.

The authors' confounding results described in Figures 6–7 are consistent with the challenges faced in other model organisms in trying to explore the function of genes in the hox cluster and the known redundancy that exists across paralogous groups and across individual clusters. Given the experimental challenges in deciphering the actual functions of individual or groups of hox genes, a discussion of the normal expression pattern of individual and groups of hox genes (and how this may change in different mutant backgrounds) could be helpful to make conclusions about likely normal function of these genes and compensation/redundancy in different mutant scenarios.

We appreciate the reviewer’s thoughtful comment. We agree that functional analyses of Hox genes are often complicated by redundancy within and across clusters. In this revision, we have included additional expression data of PG4–8 genes from the hoxba and hoxbb clusters, showing that only hoxb4a, hoxb5a, and hoxb5b are expressed in the fin buds. Although we did not analyze expression changes across mutant backgrounds in this study, we consider this an important direction for future experiments.

Reviewer #2 (Public review):

Summary:

The authors of this manuscript performed a fascinating set of zebrafish mutant analyses on hox cluster deletion and pinpointed the cause of the pectoral fin loss in one combinatorial hox cluster mutant of Hoxba and Hoxbb.

Strengths:

The study is based on a variety of existing experimental tools that enabled the authors' past construction of hox cluster mutants, and is well-designed. The manuscript is well written to report the authors' findings on the mechanism that positions the pectoral fin.

Weaknesses:

The study does not focus on the other hox clusters other than ba and bb, and is confined to the use of zebrafish, as well as the comparison with existing reports from mouse experiments.

We thank the reviewer for the thoughtful and encouraging evaluation of our manuscript. We are pleased that the strengths of our study design and clarity of writing were recognized. We also acknowledge the noted limitations, and while our focus here is on zebrafish hoxba and hoxbb clusters, we agree that future studies should expand to other hox clusters and additional models. Below, we provide individual responses to the specific points raised.

Reviewer #1 (Recommendations for the authors):

(1) Some additional expression analyses of Hoxb6/b8 etc, could be carried out to address some issues raised in the main review.

We thank the reviewer for this suggestion. In response, we performed additional whole-mount in situ hybridization analyses of PG4–8 genes from the hoxba and hoxbb clusters, including hoxb6b and hoxb8b. These experiments showed that only hoxb4a, hoxb5a, and hoxb5b are expressed in the developing fin buds, whereas hoxb6a, hoxb6b, hoxb7a, hoxb8a, and hoxb8b are not. We have incorporated these new data into the revised manuscript (Figure 7M-R), which we believe clarify why functional tests of hoxb6b and hoxb8b did not uncover specific requirements in fin development.

(2) The discussion section, particularly the more speculative section on evolutionary significance, could be reduced. Discussion of pelvic fin could be removed also, as this has not and could not be addressed with the current experimental design.

We thank the reviewer for this helpful suggestion. In line with the recommendation, we have reduced the speculative section on evolutionary significance in the Discussion to make it more concise and focused. We have also removed the discussion of pelvic fins, as these were not directly addressed by our current experimental design. We believe these changes improve the clarity and focus of the Discussion section.

(3) The conclusions on transformation to cardiac identity could be reevaluated and presented differently.

We appreciate the reviewer's insightful comment. In the revised manuscript, we have toned down our interpretation regarding a transformation to cardiac identity. Instead, we now describe the findings more cautiously, emphasizing the clear loss of fin precursors rather than a definitive acquisition of cardiac fate. We believe this revision presents a more balanced interpretation of the data.

(4) Minor typographical - I would suggest removing "Genetic Evidence:" from the title.

We appreciate the reviewer's suggestion. In accordance with this comment, we have revised the title to: "HoxB-derived hoxba and hoxbb clusters are essential for the anterior-posterior positioning of zebrafish pectoral fins".

Reviewer #2 (Recommendations for the authors):

(1) The authors mention the redundancy (between the a type and b type) of Hox clusters derived from an additional whole genome duplication in the teleost fish lineage. But, they do not refer to whether the zebrafish Tbx5 ortholog has an additional copy. This information helps the readers' interpretation of the data presented. First of all, tbx5a

suddenly appears on line 143 without introducing its relationship with Tbx5, which needs to be explained in a revised manuscript.

We thank the reviewer for highlighting this important point. In zebrafish, there are indeed two Tbx5 orthologs, tbx5a and tbx5b. In the revised manuscript, we have modified the text around line 124 to introduce tbx5a in the context of its orthology to Tbx5, ensuring that its appearance in the Results is clear to the readers.

(2) I did not readily get whether the limb/fin 'positioning' that the authors focus on in this study is 'anteroposterior' positioning, but not anything else. If it is what is meant, the word 'anteroposterior' should just be inserted at the first appearance of the word 'positioning'.

We thank the reviewer for pointing this out. Our study specifically addresses the anteroposterior positioning of paired appendages, that is, how the initial site of pectoral fin formation is defined along the anterior–posterior axis of the body. To clarify this, we have revised the text to insert the word “anteroposterior” at the first appearance of the term “positioning” in both the Abstract and Introduction (lines 26 and 53). We believe this change resolves the ambiguity and makes the focus of our study explicit.

(3) Figure 5B also shows the remarkable reduction of hoxc1a expression, which the authors do not mention at all. I wonder how this is explained and how the authors justify no remark on this throughout the manuscript.

We thank the reviewer for this insightful comment. As correctly noted, we did observe a marked reduction of hoxc1a expression in Figure 5B. However, based on our genetic analyses, we consider that the causal genes underlying the phenotype are most likely located in hoxba and hoxbb clusters. Therefore, although the change in hoxc1a expression is indeed a notable phenomenon, we did not emphasize it in the manuscript in order to maintain focus on the primary clusters responsible for the observed phenotype (lines 240-241). We agree that this point should be acknowledged, and we have now added a brief note in the Results to clarify our findings.

(4) Figure 1 consists of multiple panels (A-M) but lacks panel D.

We apologize for the oversight. We have corrected it.

(5) Line 85 - precise role -> exact role.

We have corrected it (line 95).

(6) Line 87 - the vertebrate class Actinopterygii & the class Sarcopterygii.

Thank the reviewer for pointing out. We have corrected it (line 98-99).

(7) Line 90 - homologous -> orthologous.

We have corrected it (line 102).

(8) Figure 5 - For interpretability of the data, I suggest writing 'Paralogous groups' on the top of the panels A and B, and 'Cluster' vertically on the left.

We thank the reviewer for this helpful suggestion. As recommended, we have added

“Paralogous groups” at the top of panels A and B, and “Clusters” vertically on the left side of Figure 5 to facilitate interpretation of the data.

(9) *Some subheading titles are too long. They can be shortened into 'hoxb5a and -b5b expression in pectoral fin buds are RA-dependent' instead of 'Expression patterns of hoxb5a and hoxb5b in pectoral fin buds are dependent on RA', for example.*

We appreciate the reviewer’s suggestion regarding the length of the subheading titles. In response, we have shortened the relevant subheadings in both the Results and Discussion sections to make them more concise while retaining their scientific meaning. For example, the subheading originally written as “Expression patterns of hoxb5a and hoxb5b in pectoral fin buds are dependent on RA” has been revised to “hoxb5a/b5b expression in pectoral fin buds is

RA-dependent.” Similar adjustments have been made to other subheadings throughout these sections. We believe these changes improve readability and consistency without altering the intended content.

(10) *Line 408 - why tetrapods, instead of cartilaginous fishes, which are thought of as natural in this context?*

We appreciate the reviewer’s careful reading and insightful comment. However, in response to Reviewer 1’s suggestion, we have substantially reduced the speculative section on evolutionary significance in the Discussion. As a result, this specific part of the text has now been deleted. We thank the reviewer for raising this point.

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